

Electronic Mail

Module 18

Duration: 31:13

Video: <https://www.youtube.com/watch?v=DXGMUDoZZaA>

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0:00 [Music]

0:12 Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar

0:20 University of Science and Technology Hisar harana in this lecture we will

0:27 study about email system first of all please see the contents which I will cover after covering brief introduction

0:36 about email system I shall take up email message format which will be followed by

0:44 message communication protocol SMTP in last message reading protocols

0:52 POP 3 IMAP and mapi will be explained

1:00 email which is short form of electronic mail consist of digital messages which

1:07 are sent and received using the internet it is a method of exchanging digital

1:15 messages from one sender to one or more recipient email operates across the

1:25 internet or other computer networks some early email systems required the author

1:34 and the recipient to both be online at the same time today's email systems are based on

1:43 a store and forward model email servers accept Forward

1:51 deliver and store messages neither the users nor their computers are required

1:59 to be online simultaneously an email system support

2:06 five basic functions let us see one by one composition it refers to the process of

2:15 creating messages and answers although any text editor can be

2:22 used for the body of the message the system itself can provide assistance

2:29 with with addressing and the numerous header Fields attached to each

2:36 message transfer which refers to moving messages from the originator to the

2:45 recipient reporting it has to do with telling the originator what happened to the message

2:54 was it delivered was it rejected or was it lost displaying displaying of incoming

3:03 messages is required so that people can read their email

3:11 disposition which is the final step and concerns what the recipient does with

3:17 the message after receiving it possibilities include throwing it
3:24 away before reading throwing it away after reading
3:30 saving it and few others they normally consist of two
3:38 subsystems the user agents which allow people to compose and read
3:45 email message format and sending is also taken care by user
3:52 agent the user agents are local programs that provide a command based menu
based
4:01 4 minutes, 1 second or graphical method for interacting with the email system
protocols which
4:09 supports message formatting are RFC 822 and multi-purpose internet mail
4:18 extensions that is MIME which was proposed in RFC
4:26 1341 for message reading there are three protocols that are POP 3 IMAP and
4:36 MAPI second subsystem that is message transfer agents is responsible for
4:44 actual transmission of messages from the source to destination the message
transfer agents
4:53 are typical system demons that is processes that run in the background
5:01 5 minutes, 1 second their job is to move email through the network system
protocols that support
5:08 message communication is SMTP that is simple mail transfer
5:18 protocol first we will see message format protocol for email that is
5:25 MIME the internet email message format defined by RFC
5:32 5322 with multimedia content attachments defined in RFC
5:39 2045 through RFC 2049 collectively called
5:46 multi-purpose internet mail extensions or MIME RFC
5:54 532 replaced the earlier RFC 822 in 2008 and RFC
6:03 822 in 2001 replaced RFC 82 which has been the standard for
6:11 internet email for nearly 20 years internet email messages consist of
6:19 two major sections the message header and the message body the header is
6:27 structured into fields such as from to
6:33 CC subject date and other information
6:39 about the email the body contains the message as unstructured text sometimes
6:49 containing a signature block at the end the header is separated from the
6:57 body by a blank line now first let us see message header of RFC
7:04 822 and MIME messages consist of a primitive
7:11 Envelope as described in RFC 821 with some number of header fields a
7:19 blank line and then the message body each header field logically consist of a
7:28 single line of ASCII text containing the field name a colon and a value for each

7:36 field RFC 82 was designed decades ago and does not clearly distinguish the
7:46 envelope fields from the header Fields the principal header Fields
7:53 related to message transport are listed as shown in figure
8:00 the two field gives the DNS address of the primary recipient having multiple
recipients is
8:09 also allowed the CC field gives the address of any secondary
8:17 recipients in terms of delivery there is no distinction between the primary and
secondary
8:26 recipients the BCC that is blind carbon copy field is like the CC field except
8:34 that this line is deleted from all the copies sent to the primary and secondary
8:43 recipients the next two field from and sender tell who wrote and sent the
8:52 message respectively these need not be the same a line containing received
8:59 is added by E each message transfer agent along the way this line contains
9:07 the agent's identity the date and time the message was received and other
9:16 information that can be used for finding bugs in the routing system the return
9:23 path field is added by the final message transfer agent to tell how to get back
9:31 to the sender in addition to the fields discussed just now RFC 82 messages may
9:41 also contain a variety of header Fields used by the user agents or human
9:49 recipients the most common ones are listed in figure as shown they are date
9:58 which is used to indicate date and time of the message sent reply to indicates
10:07 email address to which replies should be sent message ID indicates unique number
10:16 for referencing this message later in reply to indicates message ID
10:25 of the message to which this is a reply reference field indicates other relevant
10:34 message IDs keywords indicates user chosen
10:40 keywords and subject field indicates short summary of the message in one line
10:50 display in the early days email consisted exclusive text messages
10:56 written in English and pressed in ASCII for this environment RFC 82 explained
11:06 did the job completely it specified the headers but
11:12 left the content entirely up to the users nowadays on the worldwide internet
11:21 this approach is no longer adequate because sender want to send audio image
11:29 text video attachment with the mail a solution was proposed in RFC
11:38 1341 and updated in RFC 2045
11:45 2049 RFC 2045 to
11:50 2049 is collectively defined as RFC
11:57 532 this solution called MIME multi-purpose internet mail extensions
12:04 is now widely used MIME defines five new
12:10 message headers as shown in figure the first of these is MIME version that

12:19 simply tells the user agent receiving the message that it is dealing with a
13:26 MIM message and which version of mime it uses the content description field is an
12:35 asky string telling what is in the message the content ID field identifies
12:45 the content the content transfer encoding tells how the body is wrapped
12:53 for transmission through a network that may object to more characters other than
13:01 13 minutes, 1 second letters numbers and punctuation marks the last header
shown is really
13:09 the most interesting one and that is content type it specifies the type and
13:17 format of the content the type and subtype in this
13:23 field are separated by a slash for example video video/
13:31 MPEG various types and subtypes of content type are shown in figure various
13:39 types are text image audio video
13:46 application message and multi-art after message header it comes
13:55 message body a message body formatted according to the mime
14:01 14 minutes, 1 second specification is subdivided into parts that are organized as a
14:08 hierarchy as with any hierarchy located at the top is the root body part labeled
14:15 as R some parts contain other parts known as
14:22 descendants this figure has two direct descendant body parts label
14:29 A1 and A2 body part A1 do1 and A1 do2
14:38 that do not contain other body parts are defined as content body parts and body
14:47 part A1 and A2 that do not contain other body parts are defined as multi-art body
14:58 parts contents body part contain the actual content of the message such as the
plain
15:08 text version of the message and attachment a HTML page and so on so this
15:19 was about message format of emailing system now we will see how these
15:27 messages are transferred or communicated the message transfer system
15:35 is concerned with relaying messages from the originator to the
15:43 recipient the simplest way to do this is to establish a transport connection from
15:50 The Source machine to the destination machine and then just transfer the
15:58 message simple mail transfer protocol SMTP is an internet standard for
16:07 electronic mail transmission first defined by RFC a21 in
16:14 1982 it was last updated in 2008 with the extended SMTP editions by
16:24 RFC 5321 which is the protocol used
16:30 worldwide today SMTP is a connection oriented text based protocol in which a
16:39 mail sender communicates with a mail receiver by issuing command and
16:47 supplying necessary data over a reliable ordered data stream channel typically at
16:55 TCP connection mail processing model using

17:01 17 minutes, 1 second SMTP is as shown in figure email is submitted by a male client

17:10 muua that is male user agent to a mail server

17:16 MSA that is mail submission agent using SMTP on TCP Port

17:26 587 most mailbox provide ERS still allow submission on traditional Port

17:36 25 from there the MSA delivers the mail to its mail transfer agent

17:44 MTA often these two agents are just different instances of the same software

17:52 launched with different options on the same machine the boundary MTA has to

17:59 locate the target host it uses the domain name system DNS to look up the

18:07 male exchanger record MX for the recipient's domain which is part of the

18:15 email address on the right of at the rate sign the returned MX record contains the

18:25 name of the target host the MTA next connects to the mail

18:32 exchanger MX as an SMTP client once the

18:38 MX accepts the incoming message it hands it to a mail delivery agent

18:44 MDA for local mail delivery an MDA is able to save messages in the relevant

18:54 mailbox format again mail reception can be done

19:00 using many computers or just one an MDA May deliver messages directly

19:09 to storage or forward them over a network using

19:15 SMTP or any other means including the local mail transfer protocol

19:26 lmtpt a derivative of of SMTP designed for this

19:33 purpose mail is retrieved by end user applications called email clients using

19:43 IMAP POP 3 or IM AI protocol which we

19:49 will study in next section this was how SMTP transmission takes place now let us

19:58 see SMTP session an SMTP session consist of

20:05 commands originated by an SMTP client that is sender and corresponding

20:13 responses from the SMTP server that is receiver session is opened and session

20:21 parameters are exchanged a session may include zero or more SMTP

20:30 transactions an SMTP transaction consist of three command reply sequences as

20:38 shown in figure maale command to establish the

20:45 return address or return path or bounce address or envelope

20:52 sender this is the address to which bounce messages should be

20:59 send rcpt command to establish a recipient of this

21:06 message this command can be issued multiple times one for each recipient

21:14 these addresses are also part of the envelope data consist of a message

21:23 header and a message body separated by an empty line

21:29 now we shall discuss protocols for reading of emails interactions between email

21:38 servers and clients are governed by email protocols the three most common

21:46 email protocols for interaction or reading emails by user are post office
21:54 protocol version 3 POP 3 inter message access protocol IM map and message
22:03 application programming interface M AI most email software operates under
22:12 one of these protocols now let us see each one by one first we shall take up
22:20 pop three the basic pop three procedure is to retrieve all inbound messages is for
22:29 storage on the client delete them on server and then
22:36 disconnect the user get the email from the isp's message transfer agent
22:44 protocol POP 3 described in RFC 1939 was
22:50 originated with RFC 1081 its current specification is RFC
22:59 1939 updated with an extension mechanism RFC
23:06 249 and an authentication mechanism in RFC
23:13 1734 pop three begins when the user starts the mail reader the mail reader
23:21 calls up the ISP and establishes a TCP connection with the message trans
23:28 transfer agent at Port 110 once the connection has been
23:35 established the pop three protocol goes through three states in sequence first
23:44 authorization second transaction third update the authorization State deals with
having
23:53 the user log in the transaction State deals with the user collecting the
24:00 emails and marking them for deletion from the mail box the update State
24:08 actually causes the emails to be deleted this Behavior can be observed by
24:17 typing something like tnet mail. isp.com Port
24:26 110 here mail. isp.com represents the DNS name of isp's mail
24:36 server tnet establishes a TCP connection to Port
24:43 110 on which the pop three server listens upon accepting the TCP
24:51 connection the server sends an asky message announcing that it is present
24:59 present client starts authorization state by sending over its username and
25:07 password after a successful login the client can then send over the list
25:14 command which causes the server to list the contents of the mailbox one message
25:21 per line giving the length of that message the list is terminated by a
25:30 period Then the client can retrieve messages using the RR command and mark
them for deletion
25:40 with D E when all messages have been retrieved and possibly marked for
25:48 deletion the client gives the quit command to terminate the transaction
25:54 State and enter the update state when the server has deleted all the
26:01 26 minutes, 1 second messages it sends a reply and breaks the TCP
26:06 connection now after po3 let us see IM
26:13 map for a user with an email account which is always assessed from one PC pop

26:22 three works fine and is widely used due to its simply it and
26:30 robustness but this is always not the case many time user want to assess email
26:39 from some other computer also which is not possible in pop three this
26:46 disadvantage gave rise to an alternative delivery protocol IMAP that is internet
26:55 message assess protocol Define find in RFC
27:02 2060 the current version IMAP version 4 is defined by RFC
27:10 3501 IMAP is a newer protocol and oriented towards a connection mode of
27:19 operation the standard IM procedure is to leave messages on the server instead
27:27 of keeping copies so email is only accessible when
27:36 online IMAP does not exclude keeping copies on the client but in an inversion
27:44 of the way POP 3 works it's the servers copies that are considered the real
27:53 ones that offers an important security benefit as you won't lose your email if for some
28:02 reason client computers storage media fails an IMAP server typically listens
28:11 on well-known Port 143 IMAP provides extensive mechanisms
28:19 for reading messages or even parts of the messages with large audio and video
28:29 attachments since the working assumption is that messages will not be transferred
28:36 to the users's computer for permanent storage IMAP provides mechanisms for
28:44 creating destroying and manipulating multiple mailboxes on the
28:52 server in this way a user can maintain a mailbox for each
28:59 correspondent and move messages there from the inbox after they have been
29:08 read after IMAP it is now turn for last email reading protocol that is
29:17 mapi the message application programming interface M
29:24 AI is a proprietary email protocol of Microsoft that can be used by Outlook a
29:34 Microsoft email client software to communicate with Microsoft Exchange
29:42 which is its email server software it provides somewhat more
29:49 functionality than an IMAP protocol unfortunately as a proprietary protocol
29:57 it works only for Microsoft Outlook Exchange
30:05 interactions so friends this was about email system used in Internet
30:13 nowadays in this lecture first we covered introduction of email system
30:20 which was followed by email message format and message transmission protocol
30:30 SMTP in last message reading protocols
30:35 POP 3 IMAP and mapi were explained thank
30:42 you
30:44 [Music]
30:58 n
30:59 [Music]